ITI RESOURCE HUB - DEMONSTRATION PLAN

Trade: ICTSM | Topic: Single-Mode Optical Fiber Fusion Splicing

1. Objective: Demonstrate precision cleaving and fusion arc splicing with < 0.05dB insertion loss.
2. Tools & Equipment Required: Automatic fusion splicer, High-precision fiber cleaver, Miller strippers.
3. Step 1: Slide heat-shrink protective sleeve over the primary buffer tube.
4. Step 2: Strip outer jacket, Kevlar yarn, and 250-micron acrylate coating using thermal/mechanical stripper.
5. Step 3: Clean bare core with 99.9% pure isopropyl alcohol lint-free wipes.
6. Step 4: Cleave perpendicular at precisely 90 degrees with zero micro-cracks.
7. Step 5: Place fiber in V-grooves, initiate motorized 3-axis alignment, and execute electric arc fusion.
8. Step 6: Verify splice loss calculation on digital screen and transfer to heat-shrink oven.